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Evaluation of aeration pretreatment to prepare an inoculum for the two-stage hydrogen and methane production process

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HIGHLIGHTS

- Aerobic stress was assessed for selecting a H_2-producing inoculum.
- The biochemical hydrogen potential was tested on waste of the food industry.
- The produced inoculum was effective in H_2 and CH_4 production in a two-stage process.
- A high energy recovery as H_2 and CH_4 from waste was achieved.

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ABSTRACT

This study evaluates the effect of aeration pretreatment to prepare an inoculum for H_2 and CH_4 production in a two-stage process. Moreover, the biochemical hydrogen potential and biochemical methane potential of waste from the food industry in a two-stage process was assessed. The results confirmed the possibility of using an aerobic stress for selecting a hydrogen-producing inoculum. The inoculum was fairly stable since no hydrogenotrophic–methanogenic activity was observed in 25 days. The yields measured using glucose as substrate were of approximately 160 and 280 $N mL_{H_2} g_{COD}^{-1}$ of glucose for hydrogen and methane, respectively, which are in agreement with other studies using heat-shock for the pretreatment of the inoculum. When waste of the food industry (wheat milling) was used as substrate, a lower H_2 yield was achieved by the aerobically-pretreated inoculum if compared to heat-shock; however, when combined with methane production in a two-stage process, much higher CH_4 yield was achieved.

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1. Introduction

Over the last decade, considerable efforts have been spent to develop environmentally sustainable technologies for hydrogen production; in fact, whilst hydrogen is a clean energy carrier, nowadays it is primarily produced using non-renewable fossil fuels (Das and Veziroglu, 2001).

Biological hydrogen production can be performed through photosynthetic or dark-fermentative processes (Das and Veziroglu, 2001). Several biotechnological processes which apply different operating conditions (pure and mixed substrates, pure and mixed cultures, batch and continuous reactors, etc.) have recently been proposed for biological hydrogen production (Das and Veziroglu, 2001). The use of mixed cultures seems to be more promising as compared with pure cultures because they can be easily adapted to different substrates belonging to renewable resources or organic wastes. Moreover, well-developed and largely applied technological systems for the anaerobic treatment of wastewaters and organic solid wastes could be easily adapted for hydrogen production (Li and Fang, 2007).

Although numerous bacteria are capable of producing hydrogen, during anaerobic processes hydrogen production is usually not detected due to its rapid utilisation by other micro-organisms (Das and Veziroglu, 2001). In performing biohydrogen production tests, thus, the inoculum is usually thermally (mainly) or chemically pretreated in order to remove hydrogen-consumers whereas hydrogen-producing bacteria are capable of surviving due to their ability to produce spores (Li and Fang, 2007). However, these

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pretreatments are very expensive hindering the application of the process. Furthermore, the regrowth of hydrogenotrophic methanogens has sometimes been observed (Doungmanee et al., 2007; Spagni et al., 2010), precluding or limiting the application of continuous biohydrogen-producing systems. Spagni et al. (2010) demonstrated the regrowth of methanogens can be hindered by controlling the organic loading rate to the reactor. On the contrary, Doungmanee et al. (2007) proposed to control the growth of methanogens by repeated exposure of the sludge to high temperature (90 °C for 20 min).

Aeration has also been evaluated as pretreatment for harvesting hydrogen producing biomass (Chang et al., 2011; Ren et al., 2008; Wang and Wan, 2008; Zhu and Beland, 2006). Although aeration is also an energy-demanding process it could easily be implemented in many plants since blowers are already available in many sites (e.g., in wastewater treatment plants). Moreover, the use of aeration for controlling the process if the shift of H_2-producing to hydrogenotrophic–methanogenic conditions is observed seems to be easier than the use of thermal pretreatment. In fact, aeration requires blowers and diffusers that could be easily moved or are even already available on-site, whilst thermal treatment also requires equipment that must be resistant to heat-shock.

Since dark-fermentative processes produce volatile fatty acids (VFAs) together with hydrogen, recently a two-step process has been proposed where the produced VFAs are used as substrate for biological methane production (Antonopoulou et al., 2008a, 2008b; Chu et al., 2012; Cooney et al., 2007; Hawkes et al., 2008; Jung et al., 2013; Lu et al., 2009; Venkata Mohan et al., 2008; Xie et al., 2008a, 2008b; Zhu et al., 2008). Some studies have also demonstrated that the two-stage process for H_2 and CH_4 production can increase the methane yield in the second stage likely due to the improvement of hydrolysis in the first stage (Liu et al., 2006; Pakarinen et al., 2011).

Standardised methods for the determination of the biochemical methane potential (BMP) of organic compounds have been applied for a long time (Angelidaki et al., 2009) and similar batch tests have also been proposed for evaluating the biochemical hydrogen potential (BHP) of several substrates (e.g., Liu et al., 2013). Moreover, a batch test has recently been suggested for the determination of the hydrogen and methane potential for the two-stage fermentative process (Giordano et al., 2011).

To the best knowledge of the authors, no studies have been performed to assess aerobic pretreatment of inoculum for H_2 and CH_4 production in the two-stage process, and, in addition, to specifically evaluate the "stability" of the hydrogen-producing biomass (i.e., if hydrogen-consuming bacteria regrowth is observed). Therefore, the aims of the present study are to: (i) evaluate the use of the aerobic stress to develop a hydrogen-producing inoculum; (ii) evaluate the "stability" of the hydrogen-producing inoculum (i.e., evaluate if hydrogenotrophic methanogens regrowth is observed); and (iii) assess the BHP and BMP of organic wastes as a simulated two-stage process (i.e., biohydrogen production followed by biomethane production on the produced VFAs) by using the prepared inoculum.

2. Methods

2.1. Substrates

Glucose (Carlo Erba reagents, Italy) was used as substrate for both BHP and BMP tests. Moreover, two wastes from a typical Italian food industry (i.e., flour mill processing) were used as feedstock: (i) wheat bran of *Triticum durum* for the production of the durum wheat of the typical Italian pasta (produced in a flour mill in the province of Pavia, Northern Italy); (ii) wheat bran of *Triticum aestivum* for the production of the common wheat bread (produced in a flour mill in the province of Alessandria, Northern Italy). Before use, the wastes were stored at −20 °C.

Total solids (TS) and volatile solids (VS) of the wheat bran (both common and durum) were 88% and 84%, respectively.

2.2. Inoculum and aerobic pretreatment

Granular mesophilic sludge originating from a full-scale upflow anaerobic sludge blanket (UASB) treating waste of the food industry (potato processing) was used as inoculum for the estimation of the biochemical hydrogen and methane potential (Giordano et al., 2011). The sludge had a TS and VS content of 124 ± 20 and 83 ± 8 g L^{-1}, respectively, whilst the pH was 7.3 ± 0.5.

Aerobic pretreatment was carried out maintaining approximately 2 L of the granular sludge continuously aerated in a Plexiglas vessel using an aquarium blower and a porous stone. Aeration was maintained at approximately 100 L_{air} L_{sludge}^{-1} h^{-1}. After 2, 3, 4, 5, 6, 8, 10, 12 and 14 days of aeration, the granular sludge was used for the determination of the BHP.

The same granular sludge without pretreatment was used for the determination of the BMP.

2.3. Hydrogen production using aerobic pretreated sludge

The experiments were conducted in batch mode, in 1 L Pyrex-glass bottles maintained at constant temperature (35 ± 1 °C) in a thermostatic cabinet.

For every duration (2, 3, 4, 5, 6, 8, 10, 12 and 14 days) of the aerobic pretreament, two bottles were prepared with 120 g (drip-wet weight) of inoculum and 6.9 g of glucose. pH was corrected to 7.2 ± 0.1 by using sodium carbonate monohydrate (approximately 0.25–0.6 g). Tap water was added to reach the final volume of 600 mL, resulting to a glucose concentration of 11.5 g L^{-1} (12.305 g of COD (chemical oxygen demand) L^{-1}; 64 mmol L^{-1}).

The bottles were airtight sealed with screw caps ("Omnifit C series") which contained three valves for sampling. Anaerobic conditions were achieved by flushing the bottles with N_2 for approximately 5–10 min before the beginning of the experiments.

Two "blank" bottles were prepared as described above but without the addition of glucose. All results have been expressed as net values (subtracting the blank).

More details of the procedures, including the addition of macro- and microelements, are reported elsewhere (Giordano et al., 2011).

2.4. Stability of the aerobic pretreated inoculum for biological hydrogen production

For this experiment, three bottles (prepared as described in Section 2.3) were filled with 120 g (drip-wet weight) of aerobically (for 4 days) pretreated sludge, 6.9 g of glucose and tap water to a final volume of 600 mL. When glucose was depleted, also highlighted by the end of the biological hydrogen production, another 6.9 g of glucose was added twice consecutively (without any additional sludge).

The tests were performed at 35 ± 1 °C.

Two blank bottles were also arranged as described in Section 2.3.

2.5. Biochemical hydrogen potential (BHP) and subsequent biochemical methane potential (BMP)

The experiments were conducted according to the method for the estimation of BHP and BMP described in Giordano et al. (2011).

Three 1-L bottles were filled with 120 g of granular sludge aerobically pretreated (4 days of aeration), the substrates and tap

water to the final volume of 600 mL. At the beginning of the test, if necessary, pH was corrected to 7.2 ± 0.1 by adding $Na_2CO_3 \cdot H_2O$. The substrates tested were two wastes of the food industry (see Section 2.1) and glucose for comparison: the tests were performed using approximately (the precise amounts were measured for each bottle) 15–16 g of waste of the milling process.

The bottles were then airtight sealed using the same screw caps (Omnifit C series) described above and incubated at 35 ± 1 °C until the end of the hydrogen production.

Then, for the evaluation of the BMP, 120 g of non-pretreated granular sludge (therefore containing methanogenic bacteria) were added to every bottle in order to assess the methane production from the VFAs produced. Because of the pH decrease measured during the hydrogen production processes, 1.6–2.5 g of $Na_2CO_3 \cdot H_2O$ were added to the bottles in order to increase the pH to 7.2 ± 0.1 to optimise biomethane production.

Two blank bottles were arranged also in these experiments as described in Section 2.3.

2.6. Analyses

Gas production was measured at room temperature using the water displacement method in sodium hydroxide solution (1 M) to allow carbon dioxide removal. The gas volumes were converted to standard conditions (STP). The produced gas volume was corrected by subtracting the production of the "blanks" bottles.

Liquid and gas samples were collected through the valves of the screw caps. Gas was sampled from the headspace of the bottles using an airtight 2-mL syringe. Liquid samples were centrifuged for 15 min at 5000*g* and filtered using nylon filters (Millipore) with 0.45 μm porosity. The mixed liquor samples were used for the measurement of VFAs, glucose and pH.

Gas composition and VFAs (Acetic (HAc), propionic (HPr), *i*- (HiBu) and *n*-butyric (HnBu), valeric (HVa), caproic (HCa) and heptanoic (HHe) acids) were measured using gas-chromatographic techniques as described in Spagni et al. (2010). VFAs were expressed as COD for mass balance estimation, according to the following equivalences (unit in g of COD/g of acid) reported by Batstone et al. (2002): 1.07 for HAc, 1.51 for HPr, 1.82 for HnBu, 2.04 for HVa, 2.21 for HCa and 2.34 for HHe (note that HiBu was never detected in this study).

pH was measured using Crison probes and pH-meter. TS and VS were measured according to the Standard Methods (APHA et al., 2005).

COD of the feedstock was estimated by the oxidation of approximately 1 g of substrate with acidified dichromate according to the open reflux method described in Standard Methods (APHA et al., 2005).

Glucose was measured according to the phenol–sulphuric acid method (Dubois et al., 1956). Glucose was also expressed in terms of COD ($g_{COD_{glu}}$) using the equivalence of 1.07 g of COD/g of glucose.

2.7. Biogas yield estimation and mass balance

Several different units have been used to estimate the hydrogen production yield, which makes comparison amongst substrates difficult. In this study, the amount of electrons (as COD) were used for mass balance and yield estimation as previously proposed in Giordano et al. (2011); moreover, the estimations were carried out on the basis of the substrates applied instead of the ones converted because the amount of converted substrates cannot always be determined for complex substrates (Kraemer and Bagley, 2005). The equivalence of g_{COD}/g of VFAs, H_2, CH_4 and glucose was assumed from Batstone et al. (2002).

The gas yields (as %) were also estimated, on COD basis, relative to the initial amount of the substrates in order to evaluate the fraction of the organic matter (as electron equivalence) that can be converted to energetic gases, using the following equation:

$$Ri = \frac{Pi * fi}{CODs} \quad (1)$$

where *Ri* is the recovery for gas *i* (H_2 or CH_4; % COD/COD), *Pi* is the maximum cumulative (ultimate) H_2 or CH_4 (mol) produced, *CODs* is the amount of substrates added (g COD) at the beginning of the experiment (i.e., at the beginning of the BHP test), and *fi* is the equivalence for COD (gCOD/mol) i.e., 16 and 64 for H_2 and CH_4 (Batstone et al., 2002).

Glucose and VFAs were also expressed in terms of COD (see Section 2.6).

3. Results and discussion

3.1. Effect of the aerobic pretreatment on biohydrogen production

Continuous aeration for 2 days greatly altered the aspect of the granules since their colour changed from dark-black (typical of the anaerobic granular sludge) to bright brown–grey. Nevertheless, the aerobic pretreatment lasting 2 days was not enough to completely inhibit the hydrogenotrophic methanogens since the produced biogas presented a significant amount of methane (approximately 20%). As a result, the yield of hydrogen measured was of 67 $mL_{H_2}\ g^{-1}_{COD_{glu}}$ (Fig. 1).

The hydrogen yield obtained with aeration duration of 3–12 days was higher than the previous one (aeration of 2 days), showing values of 114—142 $mL_{H_2}\ g^{-1}_{COD_{glu}}$. Biogas was mainly composed of hydrogen with a concentration ranging from 60% to 70% and the remaining part as CO_2, whilst the methane concentration was below the detection limit of the instrument (2%), confirming the effectiveness of the aerobic pretreatment in the selection of a biomass capable to biologically produce hydrogen (Ren et al., 2008).

The prolonged sludge aeration up to 14 days caused a decrease of the measured H_2 yield (76 $mL_{H_2}\ g^{-1}_{COD_{glu}}$). However, contrary to the aerobic pretreatment lasting 2 days, no methane was measured in the biogas; moreover, no glucose was detected at the end of the test confirming that the substrate was completely depleted during the fermentative metabolism. Therefore, a prolonged aerobic treatment seemed to change the bacteria biochemical reactions to pathways not related to the biological hydrogen production. In fact it is well-known that different fermentative pathways of monosaccharides and acids can lead to different H_2 yields (Li and Fang, 2007). For example, 4 or 2 mol of H_2 can be produced from the fermentation of glucose if the end products include acetic or butyric acid, respectively; furthermore, some other fermentative end-products,

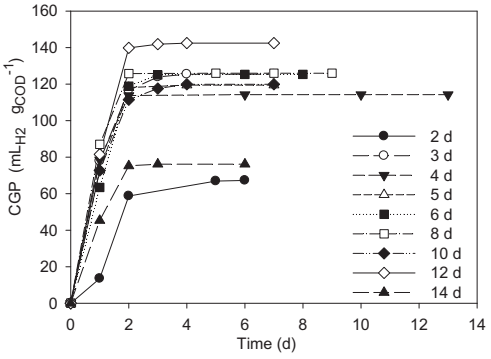


Fig. 1. Specific cumulative gas (hydrogen) production (CGP) trends measured for different durations (2–14 days) of the aeration pretreatment. Mean values of two replicates.

such as ethanol, lactate or propionate can lead to no production or, even, consumption of hydrogen (Li and Fang, 2007; Guo et al., 2010). However, since the VFAs were not monitored in this experiment, the effect of duration of the aerobic pretreatment on the fermentative pathways necessitates further investigations.

The obtained hydrogen yields of 114–142 $mL_{H_2}\ g^{-1}_{COD_{glu}}$ are equivalent to 1.0–1.2 mol of H_2 per mol of glucose ($mol_{H_2}\ mol^{-1}_{glu}$), which are included in the ranges obtained in previous studies and applying similar operating conditions but using other pretreatment (e.g., heat-shock) for the inoculum (Giordano et al., 2011; Li and Fang, 2007). These results, on the contrary, are different to those obtained by Chang et al. (2011) who achieved hydrogen production by an aerobically-pretreated inoculum that were much lower than those obtained by acid, base or heat-shock pretreatment. Moreover, the same authors (Chang et al., 2011) applied an aerobic pretreatment lasting 24 h which is 3 times lower than was required in this study to completely inhibit methanogens. However, they (Chang et al., 2011) used as inoculum the biomass obtained by an activated sludge process (collected from an aerobic wastewater treatment process) which was already acclimated to aerobic conditions and supposedly lacking methanogenic activity.

Wang and Wan (2008) also found that 24 h of aeration of sludge collected from a primary anaerobic digester were enough to select for a hydrogen-producing inoculum: however, again, the hydrogen yields obtained by aerobic pretreatment were much lower than those obtained by heat-shock. Moreover, Zhu and Beland (2006) showed that the aeration pretreatment of 30 min of sludge collected from a primary anaerobic digester did not completely inhibit methanogenic activity, confirming that longer aeration should be required to select for a hydrogen-producing inoculum. On the contrary, Ren et al. (2008) reported the repeated-aeration (i.e., by controlling dissolved oxygen concentration and the glucose feed) as the most effective pretreatment method for selecting hydrogen-producing microbial community. This study, therefore, confirming the yields obtained in previous studies using as inoculum the same anaerobic sludge but after thermal treatment (Giordano et al., 2011), supports the finding that the inoculum source and other factors, such as the duration or the intensity of the treatment, greatly affected the efficacy of the pretreatment (Ren et al., 2008; Li et al., 2010).

The results show that 3–4 days were required to obtain hydrogen-producing biomass from anaerobic granular sludge. The long time needed to inhibit the hydrogenotrophic methanogens observed in this study can be explained by the strong physical relationship amongst micro-organisms in granular sludge. In fact, thermal pretreatment has been also observed to be longer for anaerobic granules than for anaerobic suspended sludge (Alibardi et al., 2012). Furthermore, high tolerance of anaerobic granules has also been previously described for acid conditions since methanogens were found to be very active in hydrogen-producing systems operated at pH 4.5–5.5 (Spagni et al., 2010; Zhou and Ren, 2007).

3.2. Stability of the aerobic pretreated inoculum for biological hydrogen production

The experiment was carried out using the granular sludge aerobically pretreated for 4 days.

Anaerobic glucose degradation showed a total hydrogen production of 1122 ± 44 mL, resulting in a yield of 153 ± 6 $mL_{H_2}\ g^{-1}_{COD_{glu}}$ (1.3 $mol_{H_2}\ mol^{-1}_{glu}$) after 7 days (I glucose load in Fig. 2a). This yield was slightly higher than the value obtained in the previous experiment (114 $mL_{H_2}\ g^{-1}_{COD_{glu}}$ reported in Section 3.1) for the same pretreatment duration (4 days) suggesting a slight variability of the procedure.

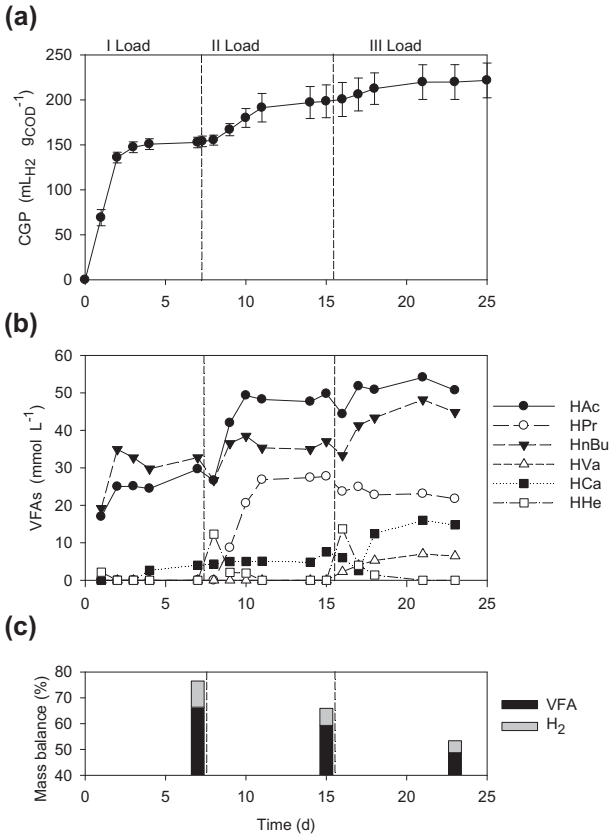


Fig. 2. Trends for (a) specific cumulative gas (H_2) production (CGP), (b) volatile fatty acids (VFAs) and (c) COD mass balance measured for repeated additions of substrate. Bars for standard errors.

With the second glucose load, a much lower yield than the first one was achieved, which reached the value of 46 $mL_{H_2}\ g^{-1}_{COD_{glu}}$. The third addition showed a H_2 yield of 23 $mL_{H_2}\ g^{-1}_{COD_{glu}}$, which was even lower than the first two (Fig. 2a). Biogas composition consisted of hydrogen with a concentration of 60–70% with the remaining percentage of carbon dioxide. No significant variation of biogas composition was observed for the following loads. Methane concentrations were always below the detection limit of the instrument for the entire duration of the experiment of 25 days. Therefore, the aerobic pretreatment stably inhibited hydrogenotrophic methanogens at least for the duration of the experiment.

The results show that although methanogenic activity was not observed, the second and the third glucose addition yielded a much lower amount of hydrogen. Chemical analysis (data not shown) revealed glucose consumption always above 94% at the end of each load (i.e., just before the second and third load) suggesting that the lower H_2 production was not caused by a limited substrate utilisation. The VFAs trends demonstrated that different metabolic pathways were involved during the subsequent additions of substrates. HnBu was the main acid produced during the fermentation of the first load of glucose, reaching the concentration of 33 mmol L^{-1} (Fig. 2b). HAc showed a slightly lower concentration (30 mmol L^{-1}) than HnBu. The other monitored acids, on the contrary, showed very low concentrations with values usually lower than the detection limit of the instrument (0.05–0.1 mmol L^{-1} depending on the acid), with the exception of HCa which showed valued up to 4.0 mmol L^{-1} just before the second glucose addition.

Acetic acid is the preferable end-product of the dark fermentative process since 4 mol of H_2 can be theoretically obtained from 1 mol of glucose (Guo et al., 2010); microbial metabolism via the butyrate pathway, instead, leads to a yield of 2 $mol_{H_2}\ mol^{-1}_{glu}$

(Guo et al., 2010). According to the theoretical yields of 2 mol_{HAc} mol_{glu}^{-1} and of 1 mol_{HnBu} mol_{glu}^{-1} from the dark fermentative reactions (Guo et al., 2010), and to the acid concentration (HAc = 30 mmol L^{-1} and HnBu = 33 mmol L^{-1}) measured at the end of the first load, the production of 126 $mmol_{H_2}$ L^{-1} can be expected (2 mol of H_2/mol of produced Hac and HnBu). Since the final liquid volume of the bottles was 0.6 L, the amount of produced H_2 is estimated in 1694 mL (126 mmol L^{-1} * 22.414 mL $mmol^{-1}$ * 0.6 L). This result confirms that the produced hydrogen is 66% (1122 mL/1694 mL * 100) of the amount forecasted by the stoichiometric reactions. The lacking 44% could be related to the involvement of other metabolic reactions that do not produce or even consume hydrogen (see below).

The second glucose addition was characterised by a very high build-up of propionic acid (Fig. 2b). Moreover, HAc concentration increased further up to 50 mol_{HAc} L^{-1}, which exceeded the HnBu values. Caproic acid, which was also detected during the first load, slightly accumulated after the second glucose load up to the concentration of 7.6 mmol L^{-1}.

Propionic acid is the product of a H_2-consuming reaction (Guo et al., 2010). The production of 28 mmol L^{-1} of HPr could be related to a consumption of an equivalent amount of H_2 (Guo et al., 2010), which resulted to a cumulative H_2 consumption of 277 mL (28 mmol L^{-1} * 22.414 mL $mmol^{-1}$ * 0.6 L). Although HAc is usually considered as the preferable end-product of the biohydrogen producing metabolism, several microbial species can also use that acid for a H_2-consuming process with a stoichiometry of 4 mol_{H_2} mol_{HAc}^{-1} (Antonopoulou et al., 2008a; Guo et al., 2010). In addition, the concomitant increase of HAc and HPr observed for the second glucose load, which resulted in a low H_2 yield, could also be justified by biochemical reactions which are neutral to the hydrogen yield such as the lactic acid fermentation (Antonopoulou et al., 2008a). However, since lactic acid was not monitored in this study, this pathway cannot be confirmed.

The third load was characterised by the formation of high-weight VFAs such as HVa (up to 7.0 mmol L^{-1}) and HCa (up to 16.0 mmol L^{-1}), although a slight further increase of HAc and HnBu was still observed. Propionic acid, on the contrary, remained almost stable suggesting that the HPr-producing (i.e., an H_2-consuming process) metabolic pathways was not involved during the third load.

It is of note that two peaks of HHe were observed just after the second and third glucose addition (Fig. 2b). However, this long-chain fatty acid was consumed during the fermentative metabolism, likely with the production of shorter chain VFAs.

The formation of long-chain fatty acids can justify (at least to some extent) the lower H_2 yields of the second and third load than the first one. In fact, having a higher molecular weight and, thus, a higher COD content, most of the electrons remained within the organic matter instead of being directed to the H_2 molecule.

Using the COD equivalences reported in Section 2.7, a COD mass balance was carried out on the glucose fed to the system and the products (H_2 and VFAs) measured at the end of each load. This mass balance revealed a limited COD closure (Fig. 2c). The COD recovery as H_2 was rather limited although in agreement with a previous study (Giordano et al., 2011), showing recoveries from 10.1% of the first load to 6.6% and 4.6% for the second and third load, respectively. The COD recovery as VFAs also decreased from 66% to 59% and 49% measured for the first, second and third glucose load, respectively. It should be pointed out that the decreasing total COD recovery (i.e., H_2 and VFAs) passing from the first (77%) to the second (66%) and the third (54%) load supports the hypothesis that other metabolic pathways could have been involved during the subsequent glucose addition. In fact, other end-products of the dark-fermentative processes (such as ethanol and lactic acid) which are not involved in H_2 producing process (and were not monitored in this study) have also been described (Guo et al., 2010).

Furthermore, several studies (e.g., Wang et al., 2008) have demonstrated that the accumulation of end-products (e.g., ethanol, acetic acid, lactic acid) of the fermentative metabolism can also significantly inhibit the biohydrogen production.

Hydrogen production caused a pH decrease from 7.7 to 4.7 after the degradation of the first amount of glucose. Because pH was not corrected, since acidic conditions should improve biological H_2 production (Li and Fang, 2007), it decreased to 4.2 with the second glucose addition and remained stable with the third one. For biological hydrogen production by dark-fermentation, the optimal pH value seems to be close to 5.5 (Li and Fang, 2007). Therefore, the reduced hydrogen yields might also be caused by a partial inhibition due to the very low pH values measured in the bottles.

3.3. Hydrogen production and subsequent methane production

All BHP and BMP tests showed very short or even no lag-phase confirming the appropriateness of the used inoculum and the applied pretreatment for hydrogen and methane production (Fig. 3).

Anaerobic glucose degradation by dark fermentation showed a hydrogen production yield of 152 ± 9 mL_{H_2} $g_{COD_{glu}}^{-1}$ (Fig. 3a) corresponding to a 163 ± 9 mL_{H_2} g_{glu}^{-1} or 1.3 mol_{H_2} mol_{glu}^{-1}. This result is completely in agreement with the tests described in the previous section.

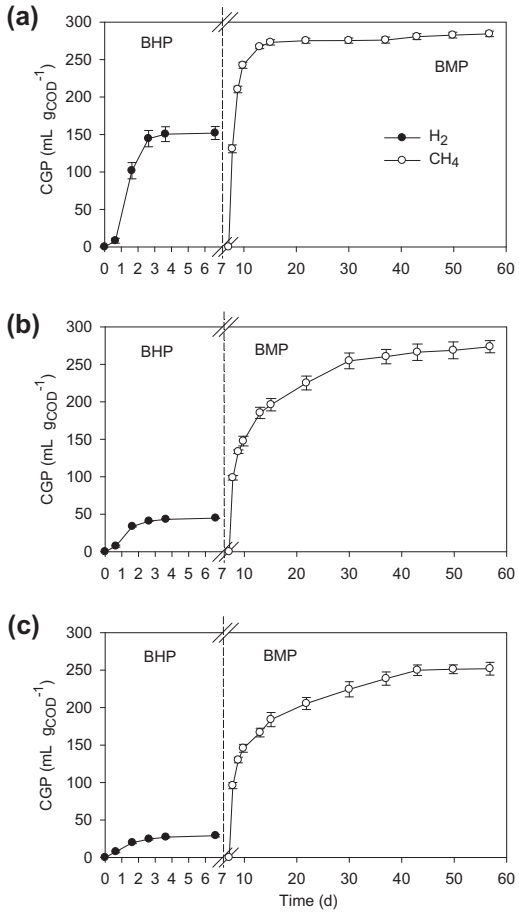


Fig. 3. Specific cumulative gas production (CGP) measured for biochemical hydrogen potential (BHP) and biochemical methane potential (BMP) of (a) glucose, (b) durum and (c) common wheat. Bars for standard errors.

The hydrogen yields obtained from wastes of the food industry by the fermentative process were, however, much lower than that obtained from glucose, showing the values of 45.8 ± 0.4 and $29 \pm 3\,mL_{H_2}\,g_{COD}^{-1}$ for durum and common wheat, respectively. These values are lower than those obtained by Hawkes et al. (2008) who reported yields of 54–91 $L_{H_2}\,kg_{vs}^{-1}$ (55 and 33 $L_{H_2}\,kg_{vs}^{-1}$ in this study, using the same unit). These values are also lower than the yields obtained by the same substrates in a previous study using the same inoculum that was, however, treated by heat-shock (Giordano et al., 2011). Therefore, the results of this study suggest that a lower H_2 yield is obtained when using an inoculum aerobically pretreated than when using thermal treatment. However, with respect to the previous study (Giordano et al., 2011), the industrial food waste was stored for approximately one year in the laboratory before use and, thus, although the substrates were relatively dry and stored at −20 °C, a partial degradation could have occurred. In addition, these differences could also be due to waste heterogeneity.

It is noteworthy that, although the BHP tests lasted for almost one week, more than 60% of hydrogen was produced in the first two days for all substrates, confirming the high activity of the H_2-producing microorganisms (Giordano et al., 2011).

As expected, VFAs were produced during the biological hydrogen production process, and promptly degraded by methanogenic bacteria (Fig. 4). HnBu and HAc were the two main VFAs produced for all substrates. However, whilst HnBu was the main fatty acid produced during the fermentation of glucose, HAc predominated when the two wastes were used as feed. In spite of this, the fermentation of the two wastes resulted in much lower H_2 yields. Furthermore, the highest HAc concentration was observed during the fermentation of the common wheat (Fig. 4c), which, however, showed the lowest H_2 yield amongst the three tested substrates. This result suggests that fermentative pathways that produce acetic acids but do not yield H_2 or even consume H_2 (Antonopoulou et al., 2008a; Guo et al., 2010) could be involved during the fermentation of the investigated waste by aerobically-treated anaerobic granules.

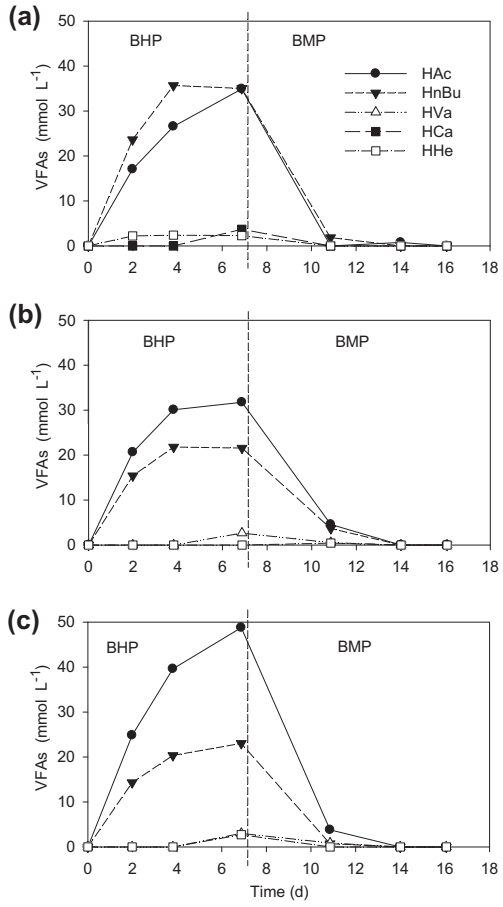


Fig. 4. Volatile fatty acids (VFAs) measured for biochemical hydrogen potential (BHP) and biochemical methane potential (BMP) of (a) glucose, (b) durum and (c) common wheat.

The other monitored VFAs always showed very low concentrations, confirming that their production occurred with extremely high load and at acidic conditions as measured during the previous experiment (see Section 3.2).

The results confirm the effectiveness of using the VFAs produced by the biological hydrogen production process as substrate for methane production via anaerobic processes (Giordano et al., 2011; Antonopoulou et al., 2008a,b).

The VFAs were completely consumed in 4–7 days under methanogenic conditions (i.e., after 11–14 days of the experiment in Fig. 4), confirming the high activity of methanogens contained in the (non-pretreated) inoculum. However, the methane production rates were very different amongst the three substrates (Fig. 3). In fact, whilst in the same period (4–7 days) more than 95% of the methane yield was obtained when glucose was used as substrate (Fig. 3a), approximately 72% and 65% of the total CH_4 yields were obtained for durum and common wheat, respectively (Fig. 3b and c). This result demonstrates that, when glucose was used as substrate, methane production was completely performed on the VFAs produced during the first H_2-producing step, whilst, during the fermentation of the two wastes, a significant amount of methane was produced from the organic matter that seemed to be still contained in the substrates (and were not released during the first H_2-producing step).

Table 1 reports the gas recovery (as defined in Eq. (1)) obtained in this study and some data from the literature for comparison. As already stated (Giordano et al., 2011; Kraemer and Bagley, 2005), the commonly used units for determination of the hydrogen yield (e.g., mol gas on mol of glucose or hexose) make difficult the comparison amongst different materials (such as wastes). Therefore, the recovery concept proposed by Kraemer and Bagley (2005) as adopted by Giordano et al. (2011) was also used in this study. A similar approach, by using electron-equivalent balance, was also adopted by Cota-Navarro et al. (2011) in evaluation the H_2 and CH_4 yields treating cheese whey powder.

The total gas (H_2 and CH_4) recovery from glucose (92.0%) approximates the mass balance closure if a biomass growth yield of 0.1 g_{COD} of biomass per g_{COD} of substrate (Batstone et al., 2002) is assumed. The check of the mass balance confirms that an increase of the biohydrogen production to the maximum experimental yield of 2.5 $mol_{H_2}\,mol_{glu}^{-1}$ (Li and Fang, 2007; Guo et al., 2010) has to result in a decrease of the methane recovery (Giordano et al., 2011).

The gas recovery also confirms that much more energy could be gained from the wastes. In fact, gas recovery comprised from 22% and 91% has been observed (Table 1) depending on the nature of the used waste, inoculum, technology and operating conditions.

It has to be highlighted, that the aerobic pretreatment of the inoculum seems to significantly improve the total gas recovery (10–18%) if compared to the same waste when the inoculum was thermally pretreated (Table 1). This improvement could be explained by substrate heterogeneity or by a partial hydrolysis of the wastes during storage. However, the improvement might also be due to better hydrolysis activity of the used inoculum. In fact, although the aim of the pretreatments of the inoculum is the inhibition of methanogens, the pretreatments can also inhibit most of the populations forming the microbial community. For example,

Table 1
Summary of the main results for biochemical hydrogen and methane potential. Values referring to this study as mean ± standard errors.

Substance	COD/VS equivalence (g_{COD}/g_{VS})	H_2 yield (mL/g_{COD})	CH_4 yield (mL/g_{COD})	Test type	Inoculum pretreatment	H_2 recovery (% COD/COD)	CH4 recovery (% COD/COD)	Total gas recovery (% COD/COD)	References
Glucose	1.07[a]	159 ± 9	284 ± 3	Batch	Aeration	10.8	81.2	92.0	This study
Durum wheat	1.19 ± 0.26	45.8 ± 0.4	274 ± 8	Batch	Aeration	3.2	78.1	81.3	This study
Common wheat	1.14 ± 0.21	29 ± 3	251 ± 9	Batch	Aeration	2.1	71.6	73.7	This study
Glucose	1.07[a]	185	267	Batch	Heat	13.3	75.5	88.8	Giordano et al. (2011)
Durum wheat	1.19	76	243	Batch	Heat	5.5	69.5	75.0	Giordano et al. (2011)
Common wheat	1.14	47	202	Batch	Heat	3.4	57.8	61.2	Giordano et al. (2011)
Mashed potatoes	1.05	177	207	Batch	Heat	12.6	59.3	71.9	Giordano et al. (2011)
Steam-peeling	1.14	134	183	Batch	Heat	9.6	52.3	61.9	Giordano et al. (2011)
Glucose	1.07[a]	320	248	Batch	Heat	22.8	70.8	83.6	Xie et al. (2008a)
Potatoes	1.2[b]	271[c]	158[c]	Batch	Heat	15.9	37.6	53.5	Xie et al. (2008b)
Potatoes	1.22	58	192	Continuous	N.A.	4.2	58.8	59.0	Zhu et al. (2008)
Wheatfeed	1.16[d]	55[e]	228[f]	Continuous	Heat	3.9	65.0	68.9	Hawkes et al. (2008)
Mix[g]	N.A.[h]	25[i]	42[i]	Continuous	None	2.8	18.9	21.8	Cooney et al. (2007)
Cheese whey	N.A.	41	310	Continuous	None	2.9	88.5	91.4	Antonopoulou et al. (2008b)
Cheese whey	1.17[j]	74[k]	231[k]	Continuous	Heat	5[k]	66[k]	71[k]	Cota-Navarro et al. (2011)
Food waste	1.1	275[l]	183[l]	Continuous	Heat	18.7	52.2	70.9	Han et al. (2005)
Food waste	1.19–1.46	27–118 [l]	267–276 [l]	Continuous	Heat	1.8–8.0	76.2–78.9	80.7–84.7	Chu et al. (2012)

[a] Unit as $g_{COD}\ g_{glu}^{-1}$.
[b] 1.2 Value assumed from Zhu et al. (2008).
[c] L gVS^{-1}.
[d] Assumed from this study.
[e] Assumed 84% of VS as in this study for wheat.
[f] Methane production was calculated theoretically.
[g] Glucose + peptone + yeast extract.
[h] Not available.
[i] Calculated from the original data that were expressed as gas mol/mol of glucose.
[j] Estimated on lactose and casein content.
[k] Estimated on e^- balance.
[l] Assumed from H_2 and CH_4 recovery.

thermal pretreatment (as other strong treatments) select for microorganisms that are able to produce spores. Since many hydrogen-producing microorganisms are able to form spores whilst methanogens are not (Li and Fang, 2007), pretreatment such as heat-shock is often selective for a H_2-producing inoculum (Li and Fang, 2007). However, the same pretreatments could also severely affect other bacteria which could have high hydrolytic activity, reducing, therefore, the capacity to degrade complex substrates such as waste or energy crops. The results of this study which show low H_2 and high gas (as the sum of H_2 and CH_4) yields (Table 1), thus, suggest that the aeration pretreatment may be less effective than other methods in selecting for a H_2-producing inoculum whilst, on the contrary, could be more suitable in preserving the hydrolytic activity which could results from a more structured microbial community. Although a specific investigation on the microbial community was not conducted in this study, Ren et al. (2008) demonstrated that different pretreatments (thermal, chemical, aeration) led to significant and diverse microbial population shift in the inoculum. Moreover, the same authors (Ren et al., 2008) also found that, amongst the pretreatments they tested, the aerobic pretreatment ensured the highest microbial diversity, supporting, therefore, the hypothesis proposed in this study.

Pakarinen et al. (2011) achieved an increase up to 64% in CH_4 yield by the addition of a thermophilic H_2-producing step before a mesophilic methanogenic stage in a two-stage process treating grass silage or maize. However, the H_2 production was always much lower than that of CH_4 (from 0.7% to 6.0% in volume) confirming the results of this study that a high CH_4 yield is obtained when H_2 production is relatively low.

4. Conclusions

The results of this study confirm the possibility of using an aerobic stress for selecting a H_2-producing inoculum. The yields measured using glucose as substrate were of approximately 160 and 280 $mL_{H_2}\ g_{COD_{glu}}^{-1}$ for H_2 and CH_4, respectively, which are in agreement with other studies using heat-shock for harvesting an H_2-producing inoculum.

When waste of the food industry was used as feed, the aerated inoculum achieved a slightly lower H_2 yield if compared to heat-shock; however, when combined with methane production in a two-stage process, much higher CH_4 yield was achieved suggesting a higher hydrolytic activity of the selected inoculum.

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